

**UNIVERSITY
OF MALAYA**

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Domain2

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<i>1</i>	<i>Ng Hwee Meng</i>
<i>2</i>	<i>Tsang Da Xin</i>
<i>3</i>	<i>Tan E Jane</i>
<i>4</i>	<i>Patrick Woo Soon Tat</i>
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1.0 Project Overview

Our project introduces a comprehensive application tailored to support Grab merchant partners in effectively managing and growing their businesses. The platform integrates various intelligent features, with the centrepiece being **Grabby**, an AI-powered chatbot that allows merchants to ask questions about their business performance such as identifying top-selling products, peak sales times or interpreting trends without needing to navigate complex dashboards.

The application also includes a **real-time insights page**, which presents business analytics through interactive charts powered by a live Google Sheets dataset. Merchants receive **instant notifications** for key updates such as performance milestones, customer engagement insights and business opportunities. The **Advice page** offers personalized, AI-generated recommendations tailored to different food categories such as strategies to boost Western cuisine sales based on order and revenue trends.

To ensure inclusivity and accessibility, the application features a **multi-language friendly user interface**, supporting multiple languages like Chinese, Malay, Tamil and Thai. This ensures that merchants from diverse backgrounds can comfortably interact with the system and fully understand their business insights. By combining intelligent tools, real-time data and language inclusivity, our platform empowers merchants to make informed decisions and grow their businesses with confidence.

2.0 Problem Statement

Grab merchant partners generate large amounts of data due to daily transactions. However, merchants lack the time, tools and expertise to effectively understand and extract meaning insights from the charts and graphs based on their sales. Without clear information on how their sales are performing, merchants struggle to improve their business performance and miss many critical opportunities by taking their business to the next level.

From what we can interpret, there is clearly a market for a solution to solve these challenges such as quickly understanding how their business is performing. They need insights to help them make informed business decisions based on trends, best-selling items and customer patterns and feedback. Without resources that could help you improve your business; merchants will slowly lose their competitive advantage. This is a lose-lose situation due to Grab losing their merchants causing less services or product selection for their customers and merchants having no profits and potentially going out of business.

To tackle the issue, we introduce Grabby, an AI chatbot that helps merchants gain insights and provide suggestions to support them. Merchants could prompt questions regarding their business such as top selling products, peak order time, graph analysis, and suggestions for business decisions. It removes the hassle for merchants to navigate the complex dashboard and it takes time to analyze and understand the analytics. With Grabby, merchants could just ask a question, for example: “When is the most popular product customer purchase and what product is it” and instantly receive all the information needed immediately. So, by utilizing AI Chabot, merchants get to focus on their business and understand how to perform better with advice personally tailored to their business. In the long run, merchants and Grab both gain profits without the risk of losing their business.

3.0 Solution Overview

To address the challenges faced by Grab merchant partners in understanding and utilizing their business data, our solution integrates an **AI-powered platform** designed to deliver actionable insights with ease and efficiency. At the heart of this solution is **Grabby**, an intelligent chatbot that allows merchants to ask questions about their business performance in natural language and receive instant, data-driven responses. From identifying top-selling items and analyzing sales trends to pinpointing peak order times and receiving strategic recommendations, Grabby simplifies the complex process of interpreting analytics. Additionally, the platform features a real-time **Insights Dashboard** with intuitive visualizations, **smart notifications** for milestones and business tips and an **AI-driven Advice Page** that offers tailored growth strategies based on the merchant's specific cuisine or product category. By transforming raw data into understandable and actionable information, our solution empowers merchants to make informed decisions quickly which ultimately helps them stay competitive, improve profitability and strengthen their partnership with Grab.

3.1 Key Features

Our application is designed with a rich set of features tailored to help Grab merchants understand and grow their business with ease. One of the core functionalities is the **Total Sales and Revenue Analysis**, which provides merchants with a comprehensive overview of their earnings and performance over time. The system also offers a dynamic **Top-Selling Products Listing**, enabling users to instantly identify their best-performing items and make data-driven decisions.

To help merchants spot trends and opportunities, the app includes **Sales Trend Visualization** tools that break down performance across daily, weekly and monthly timelines. With **Peak Order Time Insights**, merchants can optimize operations by understanding when customers are most active. Additionally, **Keyword-Based Engagement Tracking** analyses customer behaviour and preferences, giving deeper insights into what customers are looking for.

Our platform also offers **Merchant-Specific Recommendations**, personalized based on cuisine, customer patterns and historical data to guide merchants toward smart growth strategies. Through **Natural Language Interaction**, powered by Grabby the AI chatbot,

merchants can ask business-related questions and receive easy-to-understand summaries without digging through complex dashboards. Supporting a **Multi-language Friendly UI**, the app ensures accessibility for users across diverse linguistic backgrounds. Finally, **Quick Visuals through Chart Integration** makes understanding performance simple and effective with clear and attractive data visualizations.

4.0 Data Sources Used

To enable intelligent decision-making, performance tracking, and personalized responses, the system leverages a variety of structured data sources. Each dataset contributes a unique perspective, supporting both operational insight and customer engagement strategies:

1. **item.csv – Product Information and Pricing Analytics**

Contains essential data about products offered by merchants, including item names, categories, and price points. This file supports pricing analysis, product segmentation, and content personalization related to product recommendations.

2. **keyword.csv – Engagement Funnel Metrics**

Tracks user interactions across different funnel stages such as **view**, **menu**, **checkout**, and **order**. This dataset enables conversion funnel analysis and helps identify drop-off points, making it critical for optimizing customer experience and interaction design.

3. **merchant.csv – Merchant Profile Information**

Includes metadata about merchants such as business type, size, location, and date of registration. This file allows for segmentation and tailored messaging, ensuring that insights and chatbot responses are contextually relevant to each merchant.

4. **transaction_data.csv – Revenue and Order Behavior**

Records overall transaction trends, including sales volume, revenue, and time-based order patterns. This data supports the generation of performance summaries, revenue forecasting, and growth analysis.

5. **transaction_item.csv – Item-Level Sales and Performance**

Provides a granular view of sales at the item level, facilitating **bestseller identification**, inventory planning, and recommendation logic. This file is crucial for product-level insights and micro-targeted advice.

5.0 Personalization Strategy

The personalization engine in **Grabby** is designed to adapt responses based on rich contextual data, making every interaction feel relevant and value-driven for the user. Rather than offering generic insights, the chatbot dynamically adjusts its tone, recommendations, and content based on multiple real-time data factors:

- **Merchant Join Date**

Responses and performance benchmarks are calibrated based on how long the merchant has been on the platform. New merchants may receive onboarding guidance or growth tips, while long-standing ones might get deeper analytical insights and optimization suggestions.

- **City and Region**

Grabby localizes its insights by incorporating geo-specific trends. For example, it may highlight regional bestsellers, local holiday trends, or city-specific delivery performance, enhancing relevance for merchants operating in diverse locations.

- **Merchant Size and Type**

Responses are adapted depending on whether the merchant is a small boutique, a medium-sized chain, or a large enterprise. The chatbot tailors strategic recommendations based on operational scale and business model, ensuring feasibility and impact.

- **Sales History and Growth Pattern**

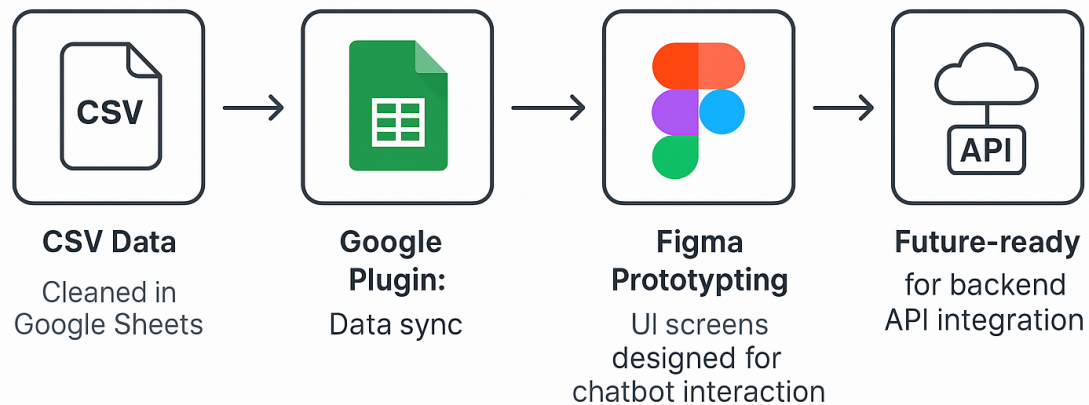
Historical transaction trends, sales seasonality, and month-over-month growth rates are analyzed to generate dynamic feedback—such as congratulating the merchant on recent spikes or recommending campaigns during slower periods.

- **Customer Behavior**

Grabby leverages engagement data such as repeat purchases, funnel performance, and product affinity to personalize marketing strategies and suggest improvements to customer retention or conversion rates.

6.0 System Architecture

This section outlines the architectural framework employed for the chatbot UI prototyping system, detailing each component from data handling to interface design, and highlighting its readiness for future API integration.



The first component, **CSV Data Management using Google Sheets**, serves as the foundation for data handling. Raw data is imported into Google Sheets in CSV format, where it undergoes cleaning, structuring, and formatting. This environment allows multiple team members to collaboratively manage the dataset, making it a live, editable source of truth. It supports real-time collaboration and ensures the data used in prototyping is accurate, up-to-date, and centralized for both designers and developers.

Next, the **Google Sheets to Figma integration** utilizes a data sync plugin, such as "Google Sheets Sync," to bind spreadsheet values directly to Figma UI components. This enables real-time synchronization between the data source and the design interface. Any changes made in the spreadsheet are immediately reflected in the Figma mockups, eliminating the need for manual updates. This integration ensures consistency between design and data, reduces the potential for human error, and significantly improves workflow efficiency.

The third component, **Figma Prototyping**, focuses on designing chatbot user interfaces that leverage real data. UI screens are developed to simulate real-time conversations and include essential chatbot components such as user inputs, chat bubbles, quick replies, and error messages. These prototypes allow for realistic interaction testing within Figma, enabling stakeholders to visualize and validate the chatbot flow. The interactive nature of these prototypes supports early user feedback and helps refine the user experience prior to backend implementation.

Finally, the architecture is **future-ready for backend API integration**, designed with flexibility and scalability in mind. The system supports seamless connection to backend services using APIs built with technologies such as Node.js, Flask, Firebase, or chatbot platforms like Dialogflow. It also allows integration with databases and AI-driven natural language processing engines. This backend-agnostic approach ensures the frontend design remains intact while enabling dynamic, real-time chatbot functionality with minimal rework. This modularity positions the system for easy transition from prototype to production.

7.0 Evaluation Criteria Mapping

To measure the effectiveness of Grabby, AI chatbot assistant, there are several criteria to evaluate so its performance can be determined. The primary goal of the evaluation is to:

1. To assess the accuracy and relevance of Grabby's responses.
2. Measure merchant's engagement and satisfaction toward Grabby.
3. Determine the impact on merchant's business when merchant uses Grabby.

There are many criteria to evaluate Grabby's usefulness and its impact:

Response Time and Accuracy

It measures the relevancy and accuracy of the answers provided by Grabby. It is important because it ensures the responses of Grabby must be reliable and furthermore personalized to the merchant's business. To measure the accuracy, it is typically best for manual reviews and feedback by merchants to determine the accuracy. As for response time, it measures time taken for Grabby to respond to merchant's prompt to ensure Grabby is fast and efficient. To measure its response time, the data will typically be reflected in system logs with time tracking.

Merchant Engagement rate and Feature Adoption Rate

It measures the frequency of merchants using Grabby to understand or ask questions regarding their business. It is important to ensure merchants use Grabby frequently so Grabby provides impact to their business. The feature adoption rate is the subset of merchant engagement to understand what advanced features are used by merchants to prove the relevancy and importance of implemented features. Features usage should be analyzed so no features are useless to merchants due to reasons like complexity or no value provided to merchants. It can be measured in usage analytics from the data collected by the Grab application.

Business Performance Uplift

It assesses its impact on the business performance of merchants with factors such as sales, efficiency and solutions taken from Grabby. The best indicator of better business performance is through sales and profits. To measure the impact of it, comparison of data between the Grabby's implementation in business and pre implementation of Grabby could be used to determine its effectiveness. Sales tracking or merchant report/feedback could be used to determine the effectiveness of it as well.

Conversation overview

It is used to ensure conversation of Grabby with merchant is natural and friendly. Grabby should be able to converse in different languages to cater to multilanguage people. The conversation between merchant and Grabby should be clear, precise and easy to understand. To measure it, feedback forms by users are best suited to determine the evaluation.

8.0 Limitations and Future Work

Grabby provides a step up to making business intelligence more accessible for Grab's merchant partner. However, during the implementation of Grabby imposes some limitations on our work. Grabby currently uses semi-live data from Google Sheets which undermines the capabilities of AI which could perform real-time data analysis and automate many analytical processes. The low accessibility of Grabby is a limitation, primary due to no voice interactions is supported. Grabby only supports text-based interactions which could be a hassle for some users.

Since the limitations are addressed, there are some preparations or improvements that needs to be made. In the foreseeable future for Grabby, API integration with Grab application backend is needed to directly integrate Grab's data collection system so that Grabby can access real-time data. Not only that, Grabby should be able to provide voice support when prompted by users, with voice support it could provide inventory and sales alert as well along with notification to notify merchants.

Currently, Grabby NLP capabilities is unable to understand context clues and slangs when interacting with users. Advanced NLP will be tailored and implemented for better interpretation of user's intent and interact. These improvements aim to make Grabby even more accessible and useful for merchants as a digital AI assistant that provides multiple functions for merchant's daily operation.

9.0 Conclusion

Grabby redefines how Grab's merchant partners engage with their business data—bridging the gap between complex analytics and actionable insights through the power of AI. By offering a natural, conversational interface, Grabby empowers merchants to easily access key performance indicators, understand trends, and receive personalized recommendations tailored to their unique business needs. The integration of real-time insights, multi-language support, and personalized advice not only simplifies operations but also drives growth, efficiency, and competitiveness in a fast-paced digital marketplace.

More than just a chatbot, Grabby is a strategic partner that transforms raw data into clear direction, helping merchants focus on what matters most: serving their customers and scaling their business. As the system continues to evolve with API integrations, voice support, and enhanced NLP capabilities, it holds the potential to become an indispensable tool across the Grab ecosystem. In doing so, Grabby supports a win-win scenario—boosting merchant success while strengthening Grab's platform and customer satisfaction.